Center for Energy-Smart Electronic Systems

Li-ion Battery Safety Research

Engineering Safety Around the Battery — Not Inside It

Meeting Agenda

- Meet the Team
- The Problem & Why It Matters
- Our Vision, Mission & Scope
- Research Framework
- What We've Built So Far
- Where We're Headed

Meet the Team

Farhan Ahamed Hameed

PhD scholar, Industrial and Systems Engineering, under **Prof. Srikanth Rangarajan.**

Research Project Assistant – Center for Energy- Smart Electronic Systems (ES2)

M.S by Research at Indian Institute of Technology Madras, India, specializing in **Battery Modeling and State Estimation** (2024 – 2025)

Area of Research: **Li-ion Battery Safety Modeling → Translational Research**

Associated with Li-ion Batteries since 2022.

Meet the Team

Dr. Srikanth Rangarajan

Assistant Professor, School of System Science and Industrial Engineering.

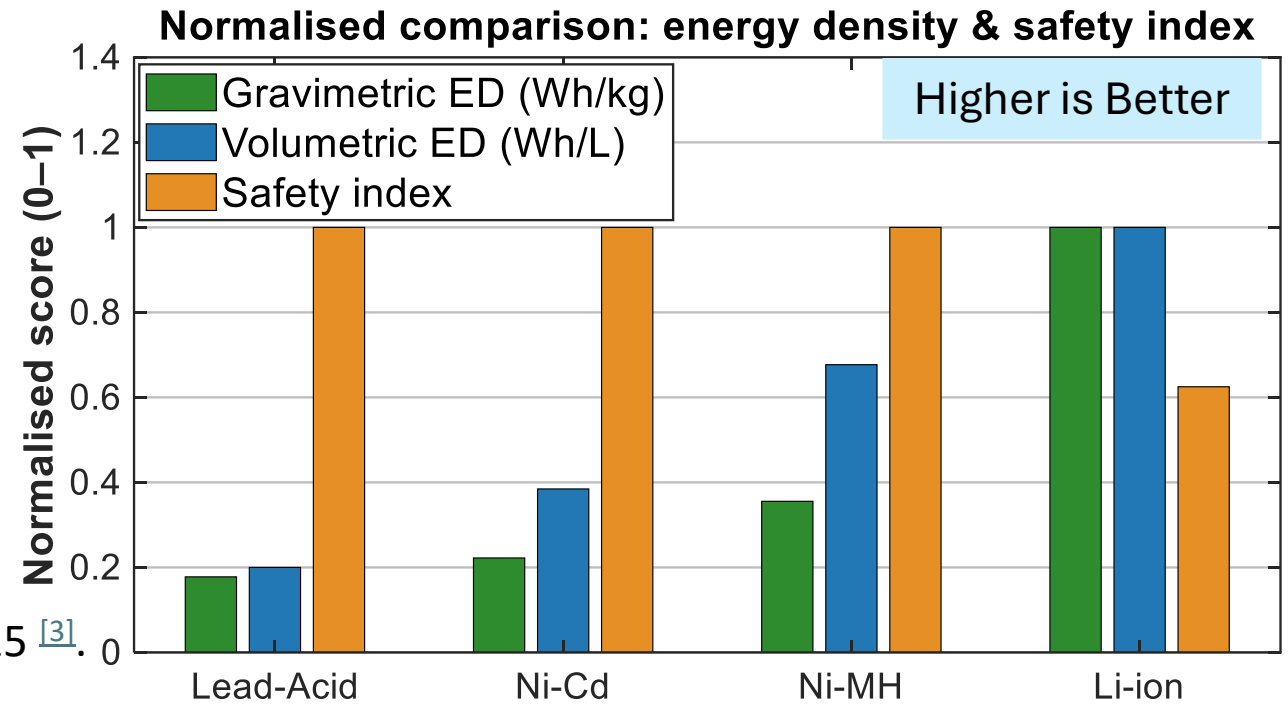
Chief Thermal Technologist, Center for Energy- Smart Electronic Systems (ES2)

M.S and PhD in mechanical engineering from the Indian Institute of Technology Madras, India specializing in Thermal Management Systems

Area of Research: Li-ion Battery Safety, Thermal Management for Electronic Systems, Data Center Cooling.

The Problem & Why It Matters

- ✓ Li-ion batteries are the current and next generation electrochemical energy storage solutions.
- ✓ From my home to your aircraft systems.
- ✓ **Safety of Li-ion Batteries is a concern.**
- ✓ **Failure: 1 in 10 million to 1 in 40 million cells [1].**
- ✓ Tesla Model 3 → **~3700 cells** per battery pack [2].
- ✓ **1 in 2000 battery pack are prone to failure.**
- ✓ Tesla sold **~1.6 million** units of Model 3 in year 2025 [3].
  → **~800 vehicles.**
- ✓ **Even though less, failure persists.**



Li-ion Battery → A ticking bomb

Make Li-ion Batteries Safer without touching what's inside them.

Make Existing Li-ion Batteries Safer.

Engineering safety around the battery, not inside it.

Our Vision, Mission & Scope

- Vision: Make Existing Li-ion Batteries Safer.
- Make sure to stay in green and green.

❑ Efficient.

❑ Effective.

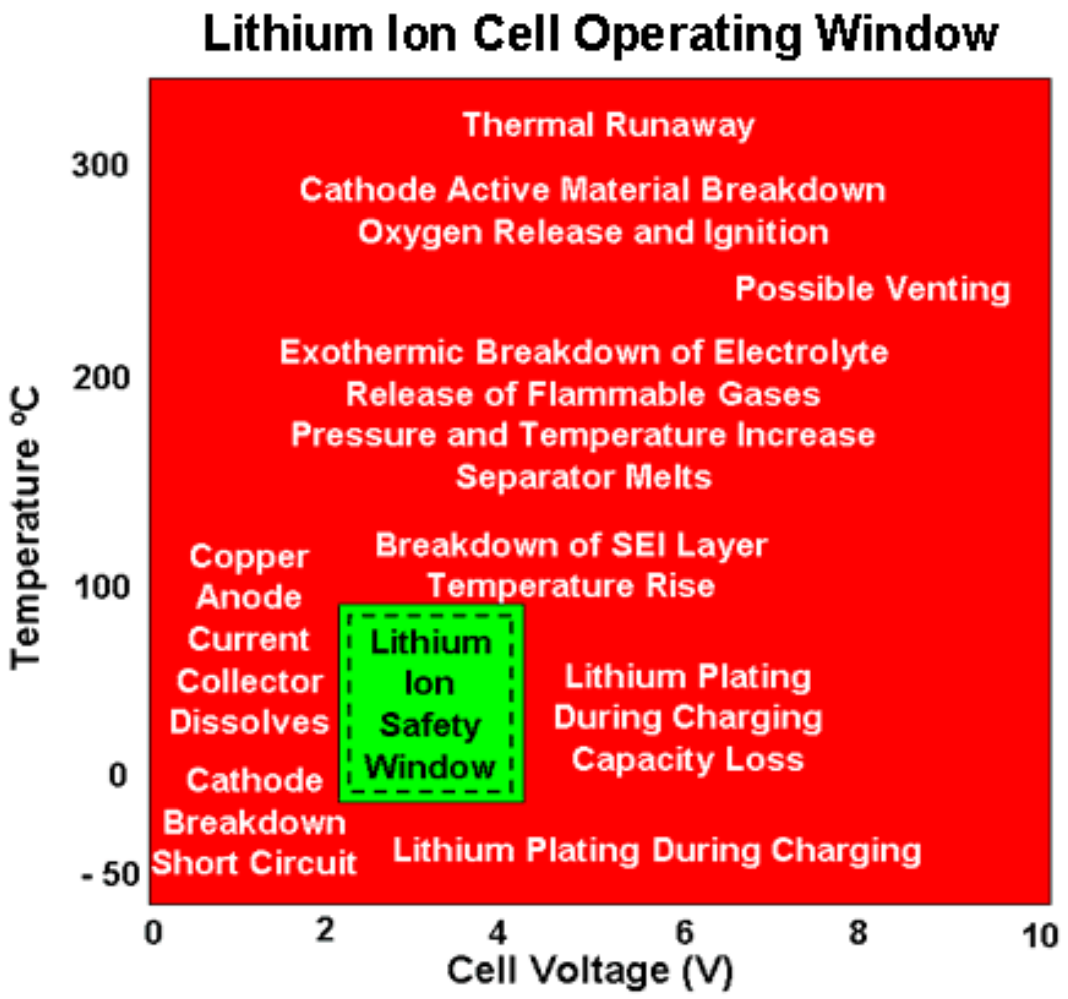
❑ Economic.

- "Doing the right thing" and "Doing it right".

Mission: To Enable,

1. Safe Design during Development &
2. Real-time Prediction during operation through

Reduced Order Modeling approach



Source: L. Lu, X. Han, J. Li, J. Hua, and M. Ouyang, "A review on the key issues for lithium-ion battery management in electric vehicles," Journal of Power Sources, vol. 226, pp. 272–288, Mar. 2013, doi: 10.1016/j.jpowsour.2012.10.060.

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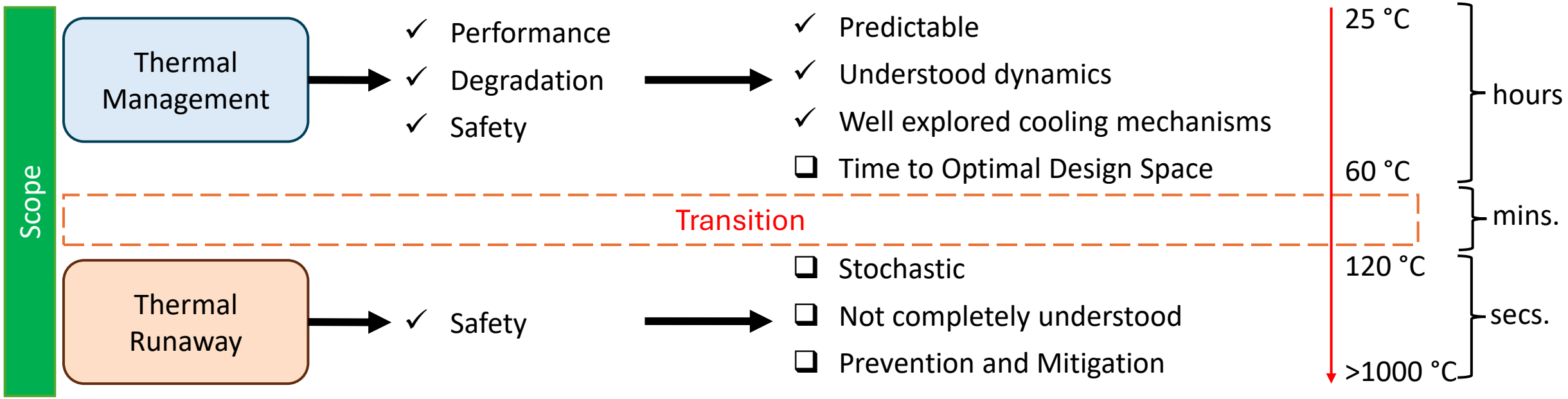
Our Vision, Mission & Scope

- Vision: To make Existing Li-ion Batteries Safer.
- Mission: To develop ROM :
  - that aids in Safe Design of Li-ion Battery Packs.
  - perform real-time predictions.



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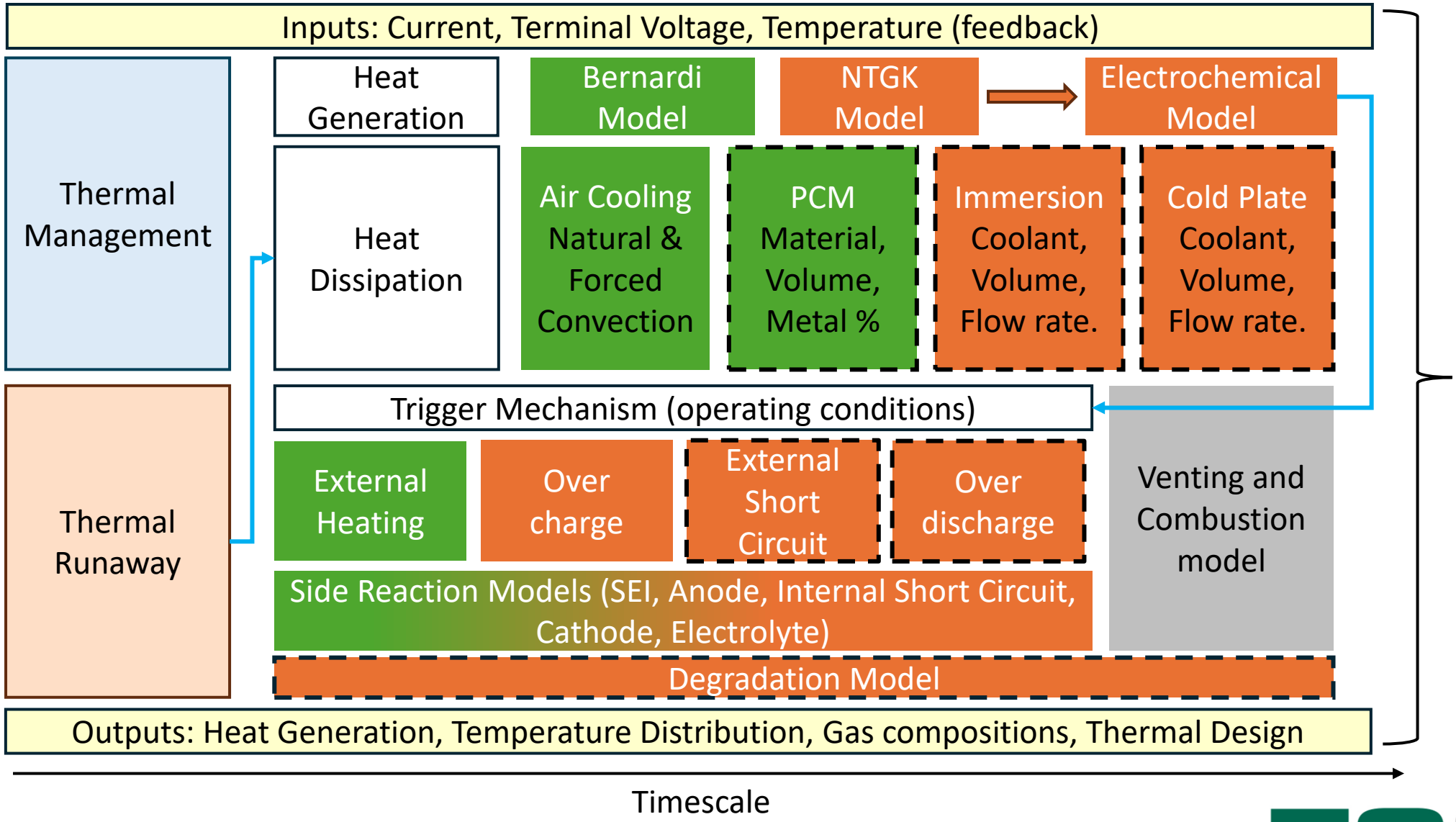
Research Framework



Project Li-therm
A ROM tool for BTMS with TR

- ✓ Transient.
- ✓ Scalable.
- ✓ Fast Design Space Exploration.
- ✓ Stochasticity.
- ✓ Optimization.
- ✓ Digital Twin.

Annotations:

- Integration
- Accomplished
- Underexplored
- Interlink

Where do we fit in Existing Landscape

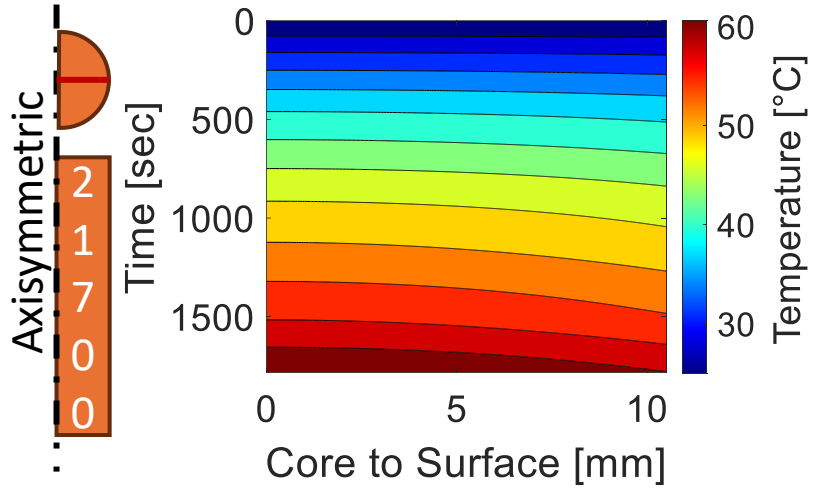
Tool	Battery	PCM	Immersion	Cold Plate	Thermal Runaway	ROM	Optimzation	Stochastic	Digital-Twin
COMMERCIAL — HIGH-FIDELITY CFD / FEM									
ANSYS	✓	✓	✓	✓	◑	✓	✓	◑	✓
COMSOL	✓	✓	✓	✓	◑	◑	✓	◑	◑
STAR-CCM+	✓	✓	✓	✓	◑	◑	✓	◑	◑
COMMERCIAL — 1D / SYSTEM-LEVEL									
Amesim	✓	✓	✓	✓	◑	✓	✓	◑	✓
GT-Lion	✓	◑	◑	✓	◑	✓	✓	◑	✓
KULI	✓	◑	✓	✓	N/A	✓	✓	◑	✓
ACADEMIA									
PybaMM	✓	◑	◑	◑	◑	✓	◑	◑	✓
DandeLiion	✓	◑	◑	◑	◑	✓	◑	◑	✓
LIONSIMBA	✓	◑	◑	◑	◑	✓	◑	◑	✓
Li-therm	✓	✓	✓	✓	✓	✓	✓	✓	✓

✓ Native / full support ◑ Partial / workaround

What We've Built So Far

Thermal Management

1D Battery (air cool)



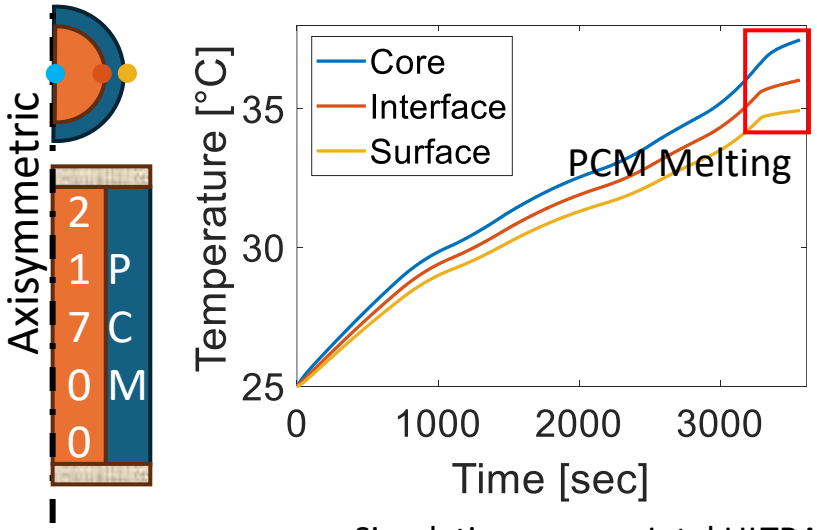
RMSE ≤ 0.5°C

MaxAE ≤ 1.5°C

200x Faster [1]

[2] Comparison to Experimental Data from Literature

1D Battery + PCM



RMSE ≤ 0.1°C

MaxAE ≤ 0.3 °C

102x Faster

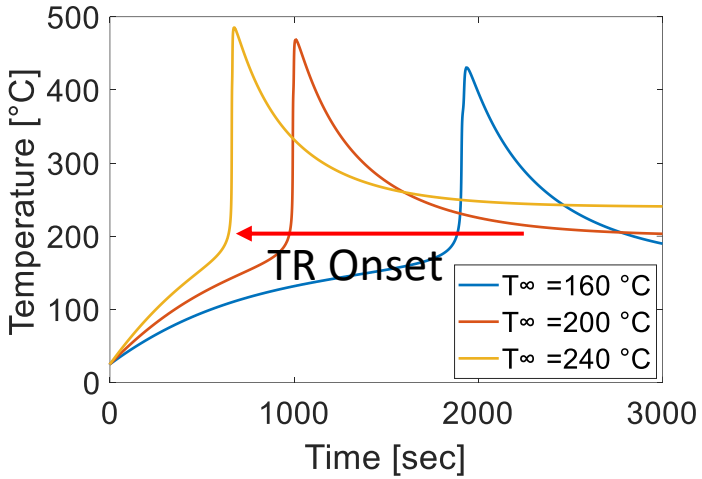
[1] Comparison to ANSYS 2D Axisymmetric

Thermal Runaway

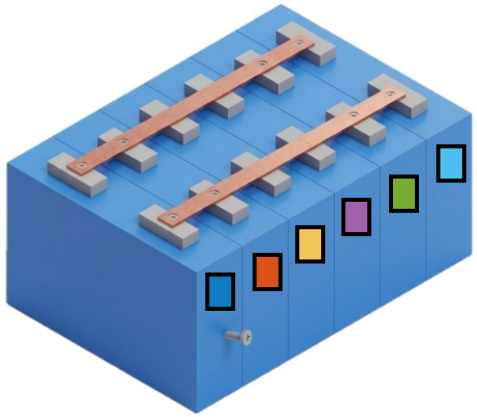
External Heating

RMSE ≤ 5 °C [2]

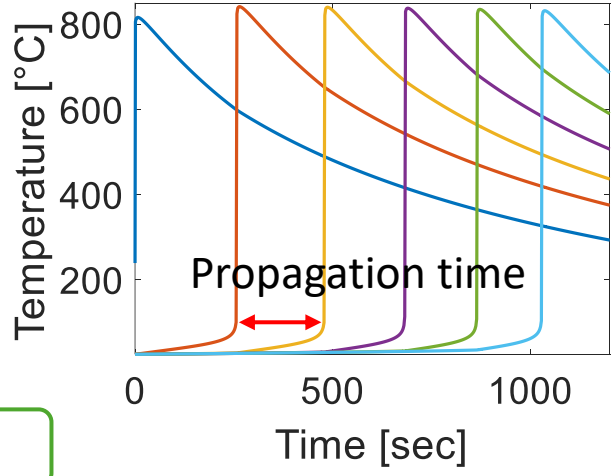
Early warning



TR Propagation



Propagation study



Simulations run on Intel ULTRA 9 285K 24 Cores @ 3.70 GHz

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Where We're Headed

Near Term Objectives (4 – 6 Months)

- ❑ Extend Thermal Management System to include Immersion and cold plate cooling mechanisms.
- ❑ Parametric Study on PCM and Immersion cooling.
- ❑ Cell interaction modeling for Pack level extension.
- ❑ Integrate Electrochemical-Thermal Model for improved accuracy.
- ❑ Li-therm: A ROM tool demonstration.

Long Term Objectives

- ❑ Thermal Runaway Modeling.
- ❑ Assist in Improving safe battery pack design pipeline.
- ❑ Digital Twin Demonstration with proposed capabilities.

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Industrial and Systems Engineering,

Battery Safety Modeling, State Estimation & Control,

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State University of New York at Binghamton,

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The greater our knowledge increases the more our ignorance unfolds.

John F. Kennedy

Thank You!

Open to Questions & Collaboration